Parameters identification of photovoltaic modules based on numerical approach for the single-diode model

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***Abstract*—This paper proposes a numerical approach for retrieving the parameters of photovoltaic (PV) panel. Based on the datasheet values, a numerical based Newton-Raphson algorithm is investigated for solving the current-voltage relation of a single diode solar PV model. To highlight the rigorous performance of our model, three modules of different technologies of PV were tested; these include the mono-crystalline, the thin-film and the multi-crystalline types. To take into account for variation in solar radiation and temperature, this model is presented under the reference and real operating conditions. The proposed model is compared with the experimental I-V and P-V data which are extracted from the manufacturer's datasheet and their performance was demonstrated. The results indicate that this model has a very satisfactory performance and it offers good compromise between accuracy and fastness.**

Keywords—Parameters extraction; Photovoltaic (PV) panel; Newton-Raphson algorithm; Accuracy

I. INTRODUCTION

From its discovery, the conversion of solar energy into electrical energy using photovoltaic modules reveals an undesirable problem. The non-linearity of the photovoltaic modules outputs, presents today a very respectful occupation of the researchers.

In order to describe the behavior of photovoltaic cells, Different models based on the current-voltage curve of a P-N junction were used. Recently, various researches on the prediction of the current-voltage characteristic curve are founded in the literature [1]. In this context, the two diodes model is known as the most accurate for representing the equivalent electrical circuit of a photovoltaic cell.

Nevertheless, additional difficulties and a longer calculation time are appended for solving the basic equation describing the two diodes model, since their parameters are defined in a nonlinear manner [2, 3]. As the most commonly used, the single diode model can be categorized into two main types [4]. The simplified four-parameter model neglecting shunt resistance by assuming it as infinite value in the equivalent electrical circuit and the five-parameter model that characterizes the current-voltage curve of the cell by maintaining the effect of the shunt resistor. The five-parameter model evaluates the photocurrent, the saturation current, the series and shunt resistors and the ideality factor of the diode.

Various methods are conceivable to determine the parameters of the equivalent electric circuit of a solar cell. However, the equation that links the different parameters which make up the current-voltage relationship is defined in a non-linear manner or implied by mathematical relationships. To solve this problem, several techniques can be considered:

Taking into account the nonlinearity relationship which cannot easily expressed of the solar panel parameters. A first approach consists to resolve this relation by the artificial neural network method [5]; this approach is used to predict the parameters of the electrical equivalent circuit of a photovoltaic cell. Another approach consists on calculating the parameters of the PV cell by the Newton-Raphson method [6-10]; that operators in an iterative manner and requires the initialization of parallel and series resistors while other parameters are defined referring to the photovoltaic modules manufacturer data.

Some analytical propositions are also suggested in this subject. In fact, an efficient analytical approach for obtaining the five parameters model of photovoltaic modules using only reference data is proposed by [11, 12]. Herein, the electrical behaviors of the photovoltaic model depending on the operation conditions of temperature and solar irradiance were described analytically. In [13, 14], various analytical solutions for approximating the parameters of the single and double diode models of a photovoltaic module have been presented, respectively. Based on manufacturer's datasheet, various numerical algorithms are investigated for solving the current-voltage equation of a single diode solar PV model [15, 16]. Other numerical methods, which are based on Newton-Raphson and Levenberg-Marquard algorithms, are proposed in [17].

In this paper, an iterative algorithm was developed to solve the nonlinearity behavior of the single-diode model in order to estimate the five parameters of the PV panel. A model based on Newton-Raphson approach is proposed for improving the accuracy of the extracted parameters that describing the

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equivalent electrical circuit of the solar cell/panel/modules. Experimental validations have been conducted in order to check the accuracy of the used model. Different technologies of the PV modules were tested for the I-V and P-V curves which are provided from the manufacturer's datasheet. Results show that the proposed model is almost agreed with the real data.

The rest of this paper is structured as follows. Section II is related to a preface on the modeling of the PV module, which describes the model statically and in general conditions of work. Section III is dedicated to the results and discussion. Finally, Section IV outlines the main conclusions.

II. MODELING OF A PHOTOVOLTAIC MODULE

Various models are used to describe the I-V curve variation of PV modules, on which we look at three characteristic points to improve the relationship between all components of equivalent electrical circuit. This one shown below was judged as the most appropriate, following an exhaustive research on this subject [18-20].

Fig.1 shows the equivalent electrical circuit of a single diode model of photovoltaic generator. It consists of a current source Iph, a diode D, a series resistance Rs and a shunt resistance Rsh.

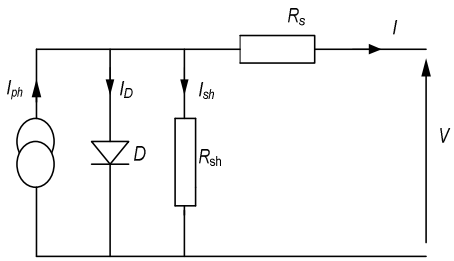

Fig. 1. Electrical circuit of a single-diode model.

The current-voltage relation of photovoltaic cell is given by:

$$I = I_{ph} - I_D - I_{sh} = I_{ph} - I_0\left(e^{\left(\frac{V+R_s I}{V_t}\right)} - 1\right) - \frac{V + R_s I}{R_{sh}} \quad (1)$$

The diode thermal voltage at reference conditions is given by:

$$V_t = \frac{AkT}{q} \quad (2)$$

where

Iph is the photo current (A); *I0* is the diode saturation current (A); A is the diode ideality factor; *k* is the Boltzmann constant (1.38×10^{-23} J/K); *q* is the electron charge (1.6×10^{-19} C); *T* is the cell temperature (K); *Rs* is the series resistance (Ω) and *Rsh* the shunt resistance (Ω).

In (1), the five parameters which defined the current-voltage relation of photovoltaic cell, vary in accordance with the solar irradiance, the cell temperature and necessary with their reference values.

All parameters of different technologies of solar panel in three characteristic points: open circuit (V_{0c}, 0); short circuit ($0, I_{sc}$); maximal power (V_m, I_m) and their temperature coefficients are illustrated in Table I.

The particulars points of current voltage, power-voltage characteristics of the solar panel used in this work are represented in Fig. 2.

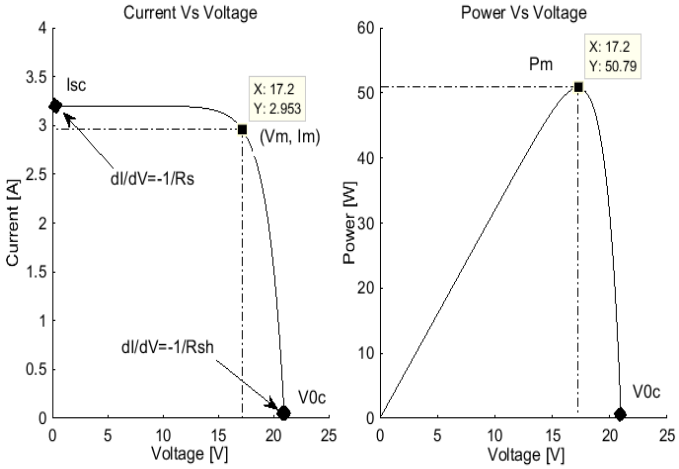

Fig. 2. Current-Voltage, Power-Voltage Characteristics and relationship between current-voltage, Rs and Rsh.

A. Static model

The general I-V characteristic of PV panel based on the single diode model is expressed by (1). In the reference conditions, we can put the current voltage relation described by (1) in the following form :

$$I = I_{phref} - I_{0ref}\left(e^{\left(\frac{V+R_{sref}I}{V_{tref}}\right)} - 1\right) - \frac{V + R_{sref}I}{R_{shref}} \quad (3)$$

where

I_{phref}, I_{0ref}, V_{tref}, R_{sref} and R_{shref} are evaluated in a particular point of current-voltage curve represented in Fig. 2, such as:

In the short circuit point: $I=I_{scref}$, $V=0$

$$I_{scref} = I_{phref} - I_{0ref}\left(e^{\left(\frac{R_{sref}I_{scref}}{V_{tref}}\right)} - 1\right) - \frac{R_{sref}I_{scref}}{R_{shref}} \quad (4)$$

In the open circuit point: $I=0$, $V=V_{0cref}$

$$I_{phref} - I_{0ref}\left(e^{\left(\frac{V_{0cref}}{V_{tref}}\right)} - 1\right) - \frac{V_{0ref}}{R_{shref}} = 0 \quad (5)$$

In the maximum power point: $V=V_{mref}$, $I=I_{mref}$

$$I_{mref} = I_{phref} - I_{0ref}\left(e^{\left(\frac{V_{mref}+R_{sref}I_{mref}}{V_{tref}}\right)} - 1\right) - \frac{V_{mref} + R_{sref}I_{mref}}{R_{shref}} \quad (6)$$

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In (3), the derivative of the current with respect to the voltage at the open circuit voltage is equivalent to the series resistance R_{sref} under standard test conditions:

$$\frac{dI}{dV}\Big|_{\substack{I=0\\V=V_{0cref}}} = -\frac{1}{R_{sref}} \tag{7}$$

In (3), the derivative of the current with respect to the voltage at the short circuit current is equivalent to the shunt resistance R_{shref} under standard test conditions:

$$\frac{dI}{dV}\Big|_{\substack{I=I_{scref}\\V=0}} = -\frac{1}{R_{shref}} \tag{8}$$

Where

I_{scref} is the short circuit current at STC

V_{0cref} is the open circuit voltage at STC

V_{mref} is the voltage at the maximum power point MPP at STC

I_{mref} is the voltage at the maximum power point MPP at STC

The above parameters are normally provided by the datasheet manufacturer's.

At MPP, the derivative of the power with respect to the voltage is equal to zero as presented in Fig. 3 and it's given by:

$$\frac{dP}{dV}\Big|_{\substack{I=I_{mref}\\V=V_{mref}}} = 0 \tag{9}$$

From (5), the photo current generated I_{phref} and the dark saturation current I_{0ref} can be obtained as:

$$I_{phref} = I_{0ref}\left(e^{\left(\frac{V_{0cref}}{V_{tref}}\right)} - 1\right) + \frac{V_{0cref}}{R_{shref}} \tag{10}$$

Substituting (10) into (4) yields:

$$I_{scref} = I_{0ref}\left(e^{\left(\frac{V_{0cref}}{V_{tref}}\right)} - e^{\left(\frac{R_{sref}I_{scref}}{V_{tref}}\right)}\right) + \frac{V_{0cref} - R_{sref}I_{scref}}{R_{shref}} \tag{11}$$

After simplification in (11), we will find:

$$I_{scref} = I_{0ref}\left(e^{\left(\frac{V_{0cref}}{V_{tref}}\right)}\right) + \frac{V_{0cref} - R_{sref}I_{scref}}{R_{shref}} \tag{12}$$

Solving this equation for I_{0ref}, results in:

$$I_{0ref} = \left(I_{scref} - \frac{V_{0cref} - R_{sref}I_{scref}}{R_{shref}}\right)\left(e^{\left(-\frac{V_{0cref}}{V_{tref}}\right)}\right) \tag{13}$$

Substituting (13) and (10) into (6) allows:

$$I_{mref} = I_{0ref} - \frac{V_{mref} + R_{sref}I_{mref} - R_{sref}I_{scref}}{R_{shref}} - \left(I_{0ref} - \frac{V_{0cref} - R_{sref}I_{scref}}{R_{shref}}\right)\left(e^{\left(\frac{V_{mref}+R_{sref}I_{mref}}{V_{tref}}\right)}\right) \tag{14}$$

Equation (9) can be rewritten as:

$$\frac{dP}{dV}\Big|_{\substack{I=I_{mref}\\V=V_{mref}}} = \frac{d(IV)}{dV} = I + \frac{dI}{dV}V = 0 \tag{15}$$

(3) is a transcendent equation, which needs numerical method to express the current and voltage. Therefore, it's rewritten as:

$$I = f(I,V) \tag{16}$$

By differencing (16), we obtain the following equation:

$$dI = df(I,V) = dI\frac{\partial f(I,V)}{\partial I} + dV\frac{\partial f(I,V)}{\partial V} \tag{17}$$

Therefore:

$$\frac{dI}{dV} = \frac{\frac{\partial f(I,V)}{\partial V}}{1 - \frac{\partial f(I,V)}{\partial I}} \tag{18}$$

From (18) and (15), we obtain:

$$\frac{dP}{dV}\Big|_{\substack{I=I_{mref}\\V=V_{mref}}} = \frac{d(IV)}{dV} = I_{mref} + \frac{\frac{\partial f(I,V)}{\partial V}}{1 - \frac{\partial f(I,V)}{\partial I}}V_{mref} \tag{20}$$

From the above:

$$\frac{dP}{dV}\Big|_{\substack{I=I_{mref}\\V=V_{mref}}} = I_{mref} + V_{mref} \times \left(\frac{\frac{-(I_{0ref} - \frac{V_{0cref} - R_{sref}I_{scref}}{R_{shref}})e^{\left(\frac{V_{mref}+R_{sref}I_{mref}}{V_{tref}}\right)}}{V_{tref}R_{shref}} - \frac{1}{R_{shref}}}{1 + \frac{(I_{0ref} - \frac{V_{0cref} - R_{sref}I_{scref}}{R_{shref}})e^{\left(\frac{V_{mref}+R_{sref}I_{mref}}{V_{tref}}\right)}}{V_{tref}R_{shref}} - \frac{R_{sref}}{R_{shref}}}\right) \tag{21}$$

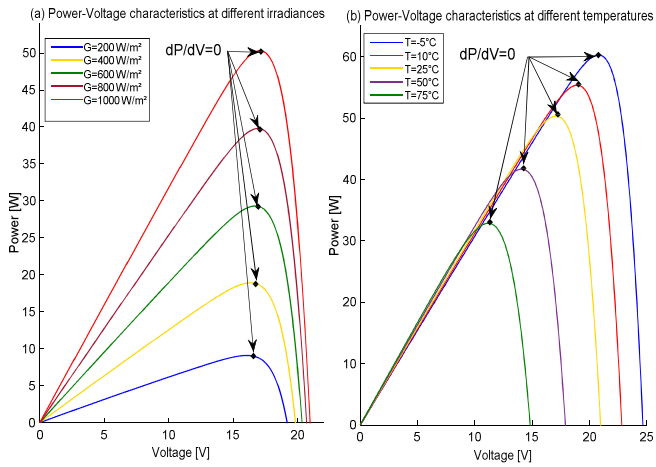

Fig. 3. P-V characteristics (a) for irradiancevariation (b) for temperature variation

There are two equations (21) and (14) with three unknowns, which are R_{sref}, R_{shref} and A_{ref}. Equation (8) can be used as the third equation. (8), (20) and (21) lead to:

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$$-\frac{1}{R_{shref}} = \frac{\dfrac{-\left(I_{0ref} - \dfrac{V_{0cref} - R_{sref} I_{scref}}{R_{shref}}\right) e^{\left(\frac{V_{mref} + R_{sref} I_{mref}}{V_{tref}}\right)}}{V_{tref} R_{shref}} - \dfrac{1}{R_{shref}}}{1 + \dfrac{\left(I_{0ref} - \dfrac{V_{0cref} - R_{sref} I_{scref}}{R_{shref}}\right) e^{\left(\frac{V_{mref} + R_{sref} I_{mref}}{V_{tref}}\right)}}{V_{tref} R_{shref}} - \dfrac{R_{sref}}{R_{shref}}} \quad (22)$$

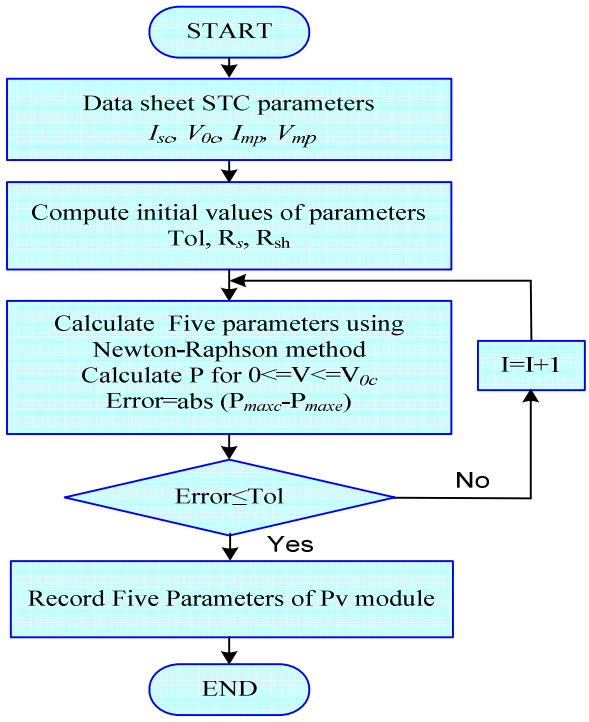

Fig. 4. flowchart of Newton-Raphson algorithm.

It's possible now to determine the five unknown parameters, which are I_{phref}, I_{0ref}, R_{sref}, R_{shref} and A_{ref} by using equations (10), (12), (14), (21) and (22).
In this work, a based Newton-Raphson algorithm is developed to extract the five parameters of the PV module which are depicted above. This method is a numerical technique which is established to estimate the five parameters of a single diode model. The main steps of the used computer algorithm for the PV model parameters estimation are shown in the flowchart presented by Fig. 4.

TABLE I. THE KEY SPECIFICATIONS OF THE DIFFERENT TECHNOLOGIES OF THE USED PV MODULES.

Characteristics	Multi-crystalline TITAN 12-50	Mono-crystalline STP275S	Thin-Film CHSM 5011T
I_{sc} (A)	3.2	9.34	1.140
V_{0c} (V)	21	38.5	170
P_m (W)	50	275	130
I_m (A)	2.9	8.85	0.965
V_m (V)	17.2	31.1	134.7
α	0.1%/°C	0.060%/°C	0.05%/C
B	-0.38%/°C	-0.34%/°C	-0.31%/C
γ	-0.47%/°C	-0.41%/°C	-0.27%/C

B. *Parameters processing in general conditions*

The dynamic behavior of the PV cell/ panel or module can be considered when slowly varying the solar irradiance and the temperature during a long period of time. Then, the electrical quantities will be evaluated as function of time.

The PV module parameters at different temperature and irradiation conditions are defined by the following way [8]:

✓ The ideality factor:

$$V_t = V_{tref}\left(\frac{T}{T_{ref}}\right) \quad (23)$$

where

T and T_{ref} are the cell and reference temperatures respectively.

✓ The saturation current of the diode :

$$I_0 = I_{0ref}\left(\frac{T}{T_{ref}}\right)^3 e^{\left(\frac{E_g N_s}{V_{tref}\left(1-\frac{T_{ref}}{T}\right)}\right)} \quad (24)$$

where

E_g is the band gap energy of semi conductor (1.12 eV for Si) and N_s is the number of series solar cells composed the PV module

✓ The photo-current:

$$I_{ph} = \left(G/G_{ref}\right)\left(I_{phref} + \alpha_{Isc}(T - T_{ref})\right) \quad (25)$$

where
G and G_{ref} are respectively the solar irradiances at real conditions and at Standard test conditions (STC) (W/m²); α_{Isc} is the temperature coefficient of short circuit current (A/°C).

✓ The serie resistance:

$$R_s = R_{sref} - \left[\left(\frac{V_t}{I_0}\right) e^{(-V_{0c}/V_t)}\right] \quad (26)$$

where
V_{0c} is the voltage which represents the open circuit point; it is expressed by the following relation:

$$V_{0c} = V_{0cref} - \beta(T_{ref} - T) + V_t \ln(G/G_{ref}) \quad (27)$$

✓ The shunt resistance:

$$R_{sh} = R_{shref}(G/G_{ref}) \quad (28)$$

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- ✓ The short circuit current:

$$I_{sc} = I_{scref}\,(G/G_{ref}) + \alpha_{Isc}(T - T_{ref}) \quad (29)$$

- ✓ The maximum power point current :

$$I_m = I_{mref}\,(G/G_{ref}) \quad (30)$$

- ✓ The maximum power point voltage:

$$V_m = V_{mref} - \beta(T_{ref} - T) \quad (31)$$

where

β is the temperature coefficient of the open circuit voltage (V/°C).

III. Results and Discussion

The performance of the model described in this work is validated by measured parameters of different technologies of PV modules. The I-V and P-V data are extracted from the manufacturer's datasheet. Three different types of technologies are used which comprise the multi-crystalline, the mono-crystalline and the thin-film.

For accurately tested the used model, the solar irradiance is varied from 200 to 1000 W/m². Fig. 5 shows the I-V and the P-V characteristics for TITAN 12-50 for different levels of irradiance. The I-V and P-V curves for STP275S for different levels of irradiance are shown in Fig. 6. The CHSM 5011T is also tested for different levels of irradiance and the I-V, P-V characteristics of this type are presented in Fig. 7. For all types of modules, we can be seen that the curves obtained by this model are very agreed to the experimental data.

In order to quantify the accuracy and the goodness of the proposed model over current-voltage and power-voltage characteristics, a mean relative error analysis was applied. In this part, five different test conditions of solar irradiation are taken account. The mean relative errors E_X values are obtained using the following equation as follows:

$$E_X(\%) = 100\frac{I_{exp} - I_{mod}}{I_{mod}} \quad (32)$$

where

I_{mod}, I_{exp} are the extracted and the experimental parameters, respectively and X= $[V_{0c}, I_{sc}, V_m, I_m]$ is the all parameters set to compare.

It's found that the proposed model agree very well with the experimental values with some uncertainty at different levels of irradiance.
The five different parameters of this model for the three used technologies are illustrated in Table II.

Table III shows the mean relative error E_x for the three characteristic points (*V0c, Isc,* and *MPP*) respectively computed with the proposed model for the different PV modules.

TABLE II. PARAMETERS EXTRACTED FOR DIFFERENT SOLAR IRRADIANCE LEVELS OF DIFFERENT PV MODULES.

Parameters	Multi-crystalline TITAN 12-50	Mono-crystalline STP275S	Thin-Film CHSM 5011T
G= 200 W/m²			
I_{ph} (A)	0.640960	1.868125	0.228049
I_0 (A)	1.81e-08	4.18e-07	2.89e-09
R_s (Ω)	0. 093955	0.060645	0.181936
R_{sh} (Ω)	606.454823	4548.411171	4245.183760
A	1.109202	2.276053	1.673979
G= 400 W/m²			
I_{ph} (A)	1.281920	3.736249	0.456098
I_0 (A)	1.81e-08	4.18e-07	2.89e-09
R_s (Ω)	0.054047	0.060645	0.181936
R_{sh} (Ω)	303.227411	2274.205586	2122.591880
A	1.109202	2.276053	1.673979
G= 800 W/m²			
I_{ph} (A)	2.563840	7.472498	0.912195
I_0 (A)	1.81e-08	4.18e-07	2.89e-09
R_s (Ω)	0.122653	0.060645	0.181936
R_{sh} (Ω)	151.613706	1137.102793	1061.295940
A	1.109202	2.276053	1.673979
G=1000W/m²			
I_{ph} (A)	3.204800	9.340623	1.140244
I_0 (A)	1.81e-08	4.18e-07	2.89e-09
R_s (Ω)	0.135652	0.060645	0.181936
R_{sh} (Ω)	121.290965	909.682234	849.036752
A	1.109202	2.276053	1.673979

TABLE III. MEAN RELATIVE ERROR FOR DIFFERENT PV MODULES.

Error	Multi-crystalline TITAN 12-50	Mono-crystalline STP275S	Thin-Film CHSM 5011T
G=200W/m²			
E_{V0c} (%)	2.7900	0.0023	0.3200
E_{Isc} (%)	0.1600	2.4100	0.8400
E_{Im} (%)	0. 093955	0.065	3.421
E_{Vm} (%)	1.9873	2.4800	2.3190
G=1000W/m²			
E_{V0c} (%)	0.6300	0.3600	0.6500
E_{Isc} (%)	0.8000	1.0600	0.4400
E_{Im} (%)	0.9735	2.2200	2.9000
E_{Vm} (%)	1.5800	2	1.2600

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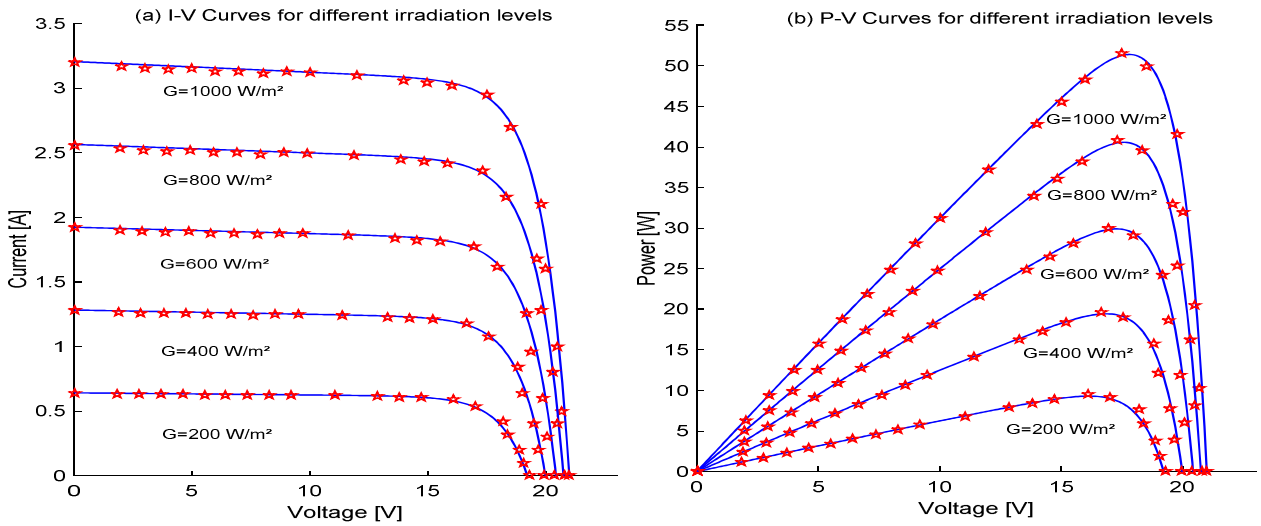

Fig. 5. (a) I-V, (b) P-V characteristics for different irradiation levels (Multi-crystalline TITAN 12-50).

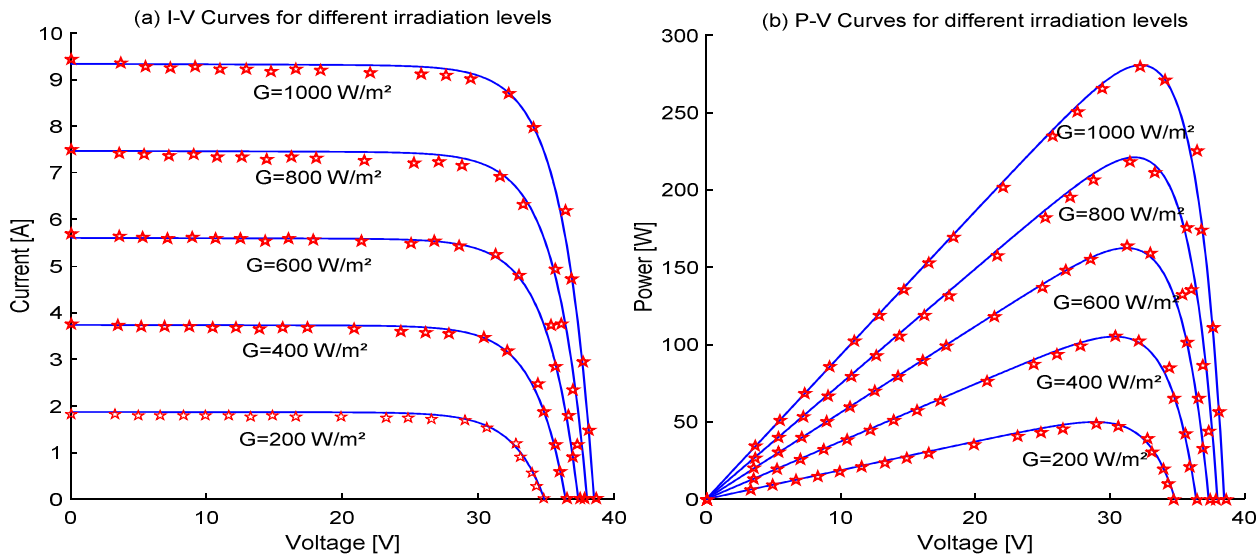

Fig. 6. (a) I-V, (b) P-V characteristics for different irradiation levels (Mono-crystalline STP275S).

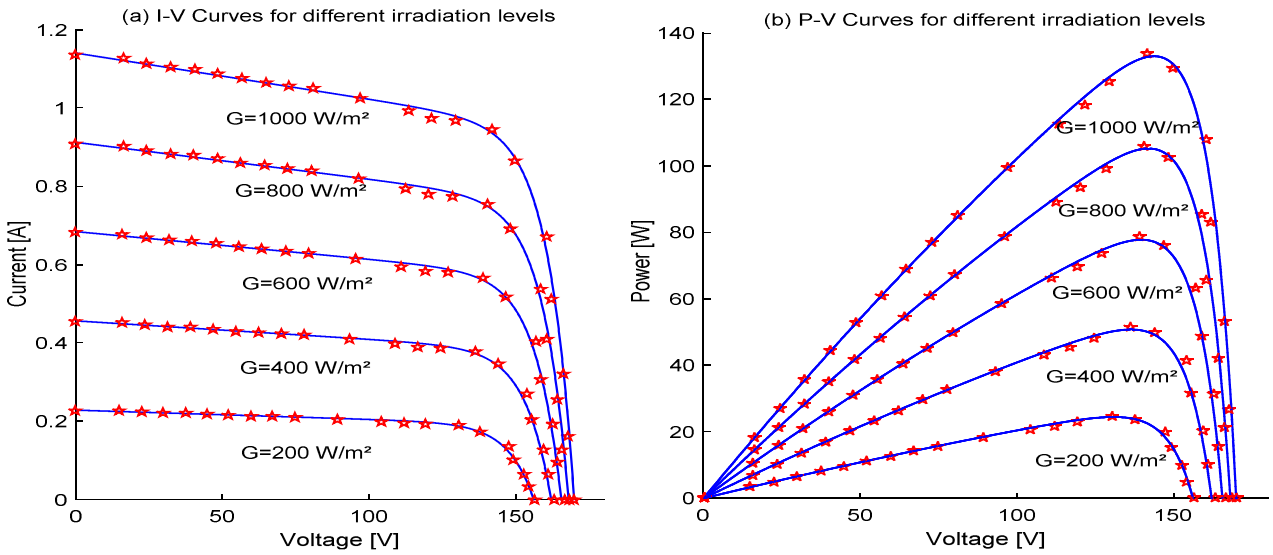

Fig. 7. (a) I-V, (b) P-V characteristics for different irradiation levels (Thin-Film CHSM 5011T).

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The recapitulation of this work proves that the proposed model based on Newton-Raphson algorithm is very appropriate to describe the behavior of the PV modules. The current-voltage and power-voltage curves of this model are exactly corresponds to the experimental values for the different levels of solar irradiance. After comparison, the obtained results indicate the durability, the accuracy and the satisfactory performance of the proposed model to significantly reduce the error in the extracted parameters for solar panels.

IV. Conclusion

This paper presents accurately a single diode parameters estimation model based on numerical approach. A numerical based Newton-Raphson algorithm model has been mainly used to determine the five parameters of the PV panel. Furthermore, the developed model has been applied to improve the performance of the parameters extraction. This approach was designed to adapt to the variations in solar radiation and to ensure a more acceptable I-V and P-V characteristics. Convincing results since the formulated approaches were considered as an optimization problem for determining the five dominating parameters of the single diode model. The performed model was experimentally validated; their accuracy and performance were discussed. Furthermore, it has been proven that the used approach has a more satisfactory performance referring to the mean relative error of identified parameters for different technologies of PV modules.

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